Міністерство освіти і науки України

Київський національний університет імені Тараса Шевченка

Фізичний факультет, кафедра загальної фізики

ДОДАТКОВІ МАТЕРІАЛИ

до дисертації на здобуття наукового ступеня

доктора філософії зі спеціальності «104 – фізика та астрономія»

«Характеризація дефектів фотоелектричних перетворювачів

на основі вольт-амперних вимірювань та глибокого навчання»

Розділ 3: Вплив залізовмісних дефектів на характеристики та параметри фотоелектричного перетворення кремнієвих сонячних елементів

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м. Київ – 2025

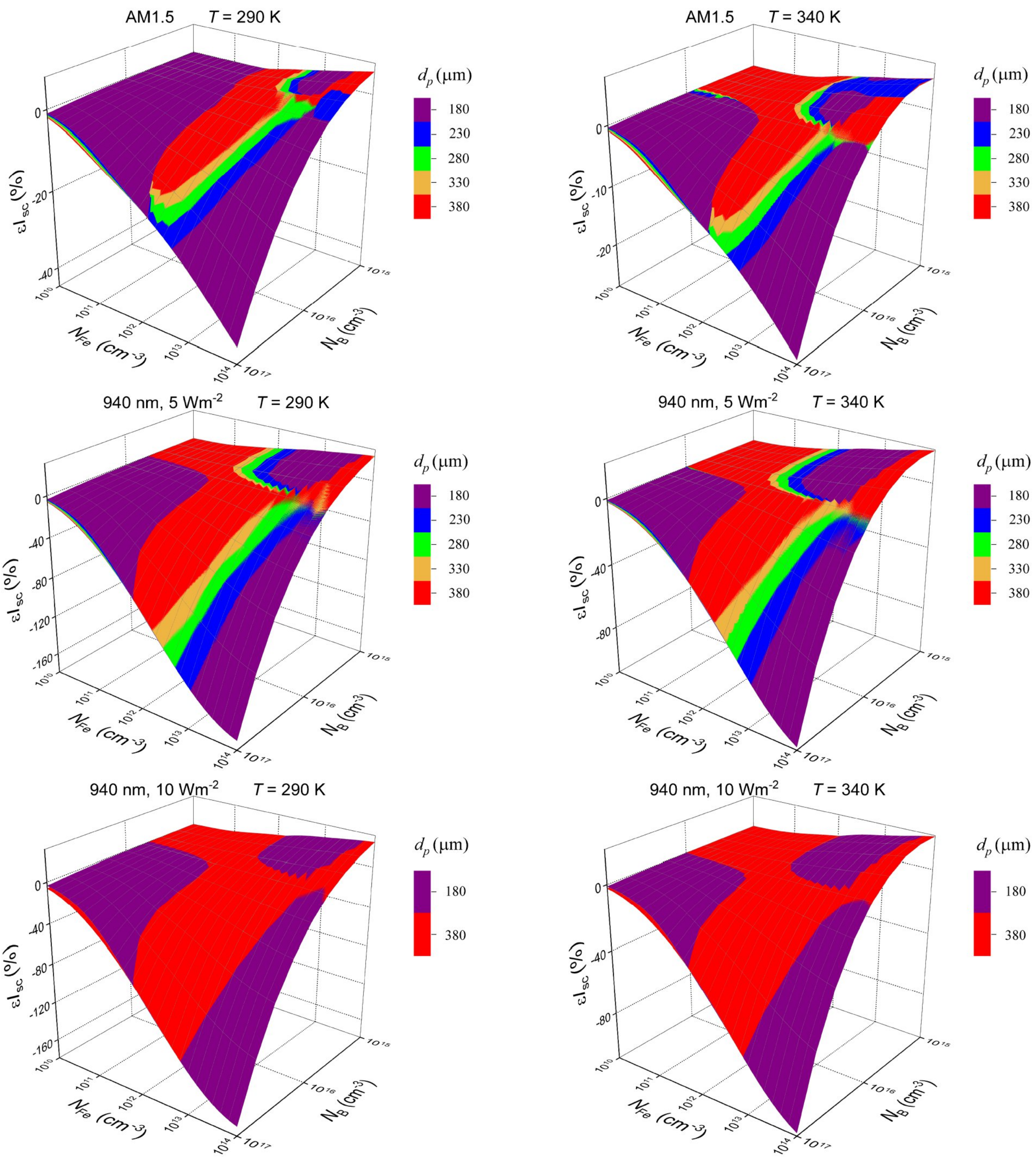

Fig.S3.1. Relative changes in short-circuit current caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level for SC with different base depth. T, K: 290 (left panels), 340 (right panels).

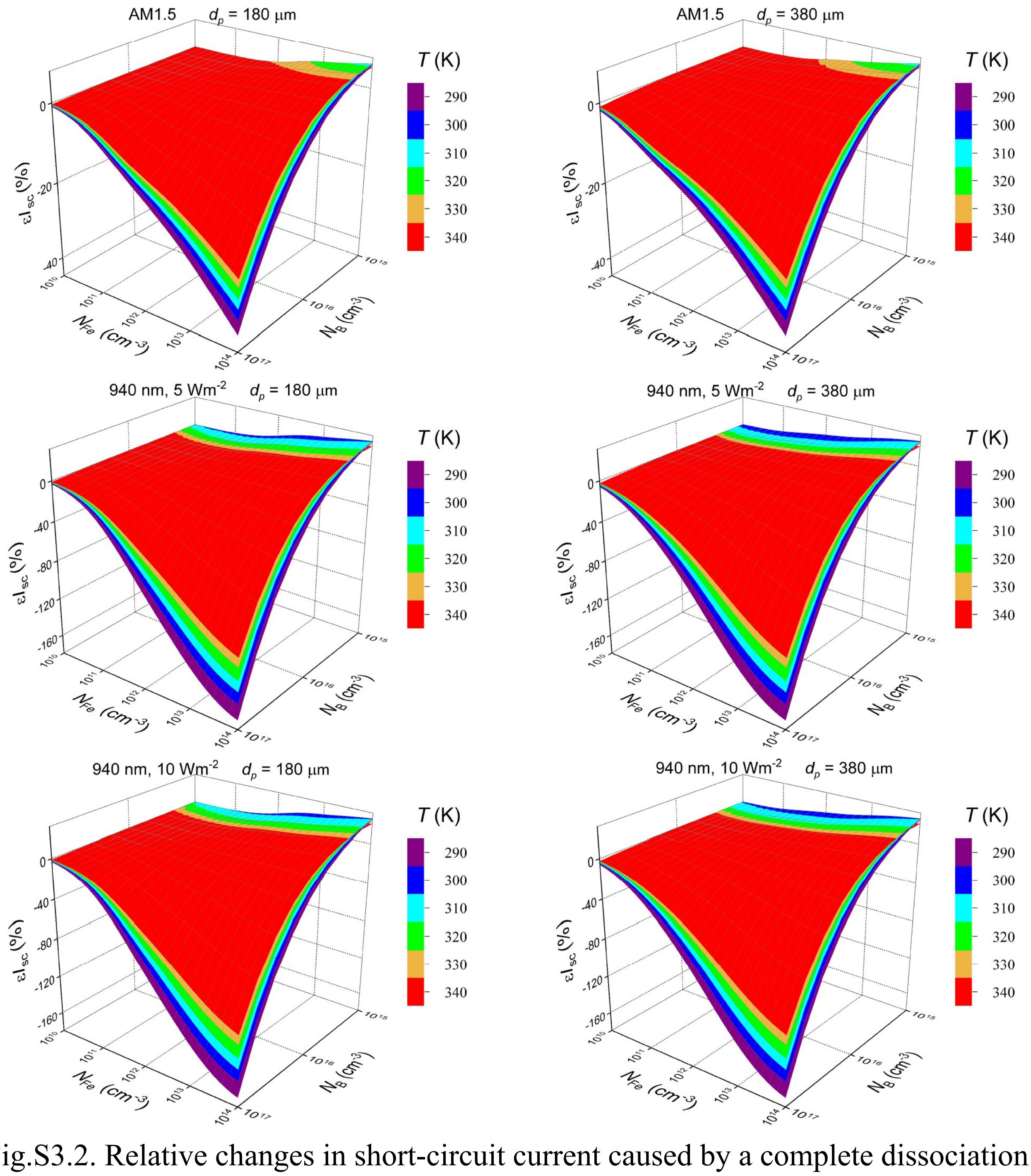

Fig.S3.2. Relative changes in short-circuit current caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level at different temperatures. d_p, μm: 180 (left panels), 380 (right panels).

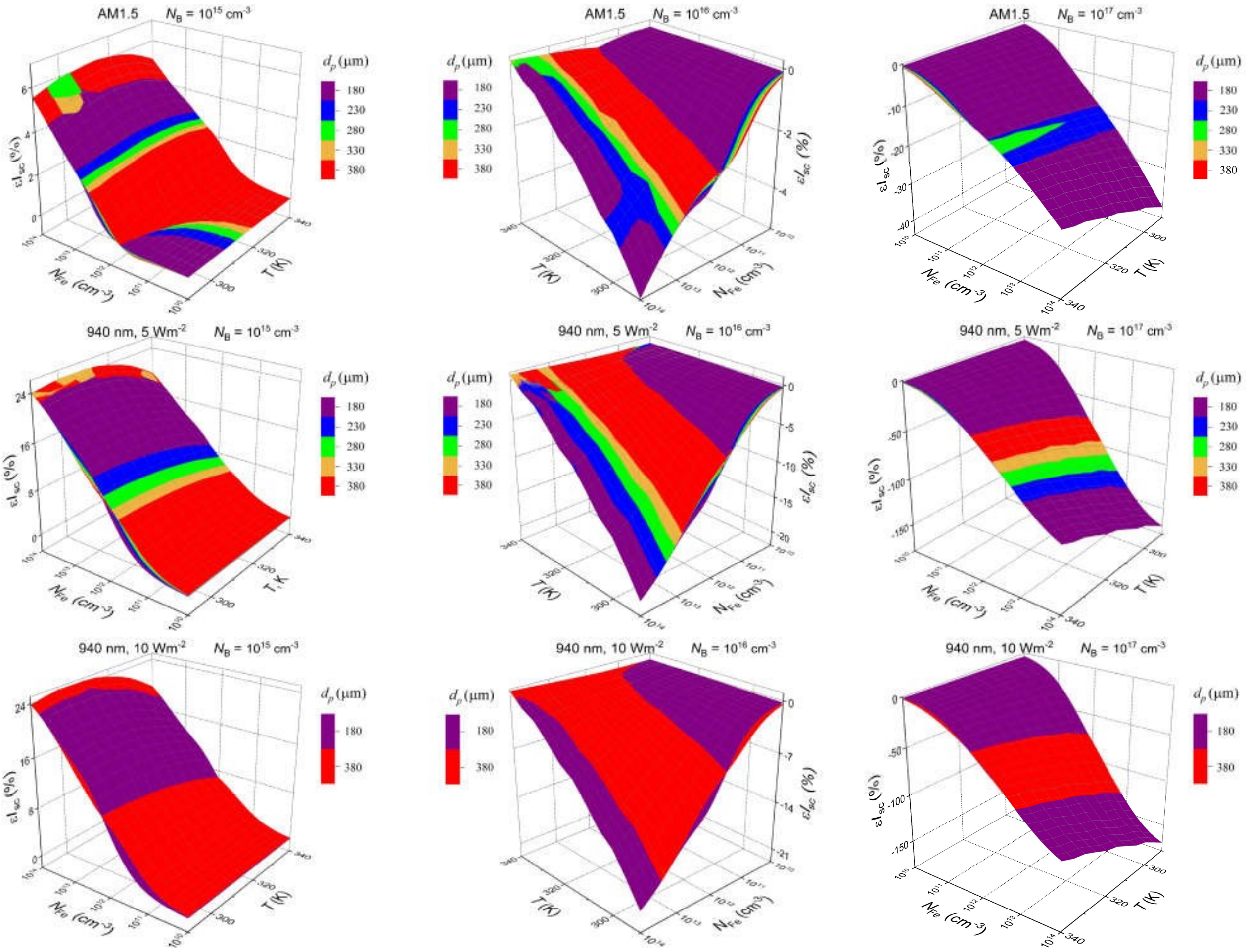

Fig.S3.3 Relative changes in short-circuit current caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and temperature for SC with different base depth. N_B, cm^{-3}: 10^{15} (left panels), 10^{16} (middle panels), 10^{17} (right panels).

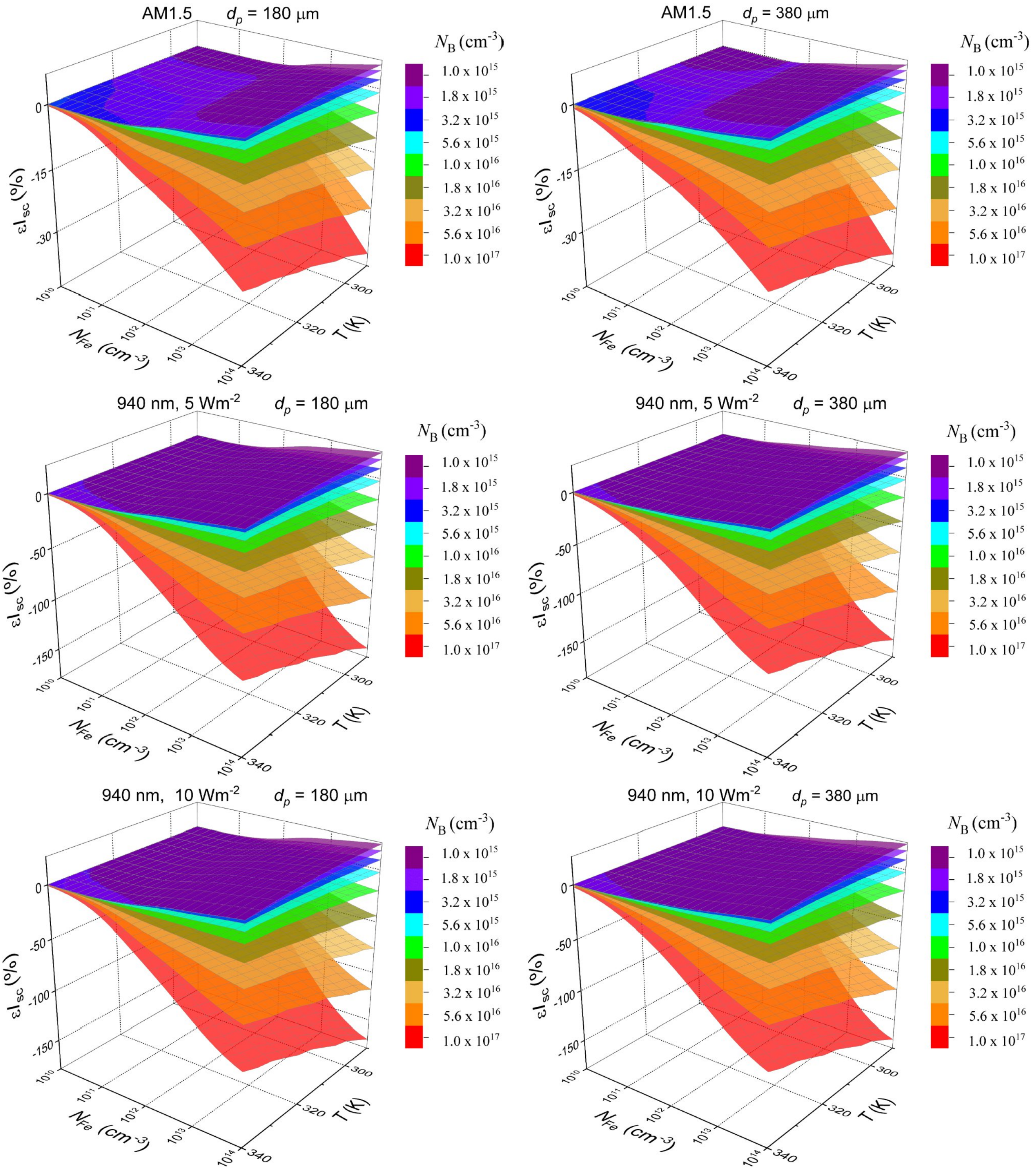

Fig.S3.4. Relative changes in short-circuit current caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and temperature for SC with different base doping level. d_p, μm: 180 (left panels), 380 (right panels).

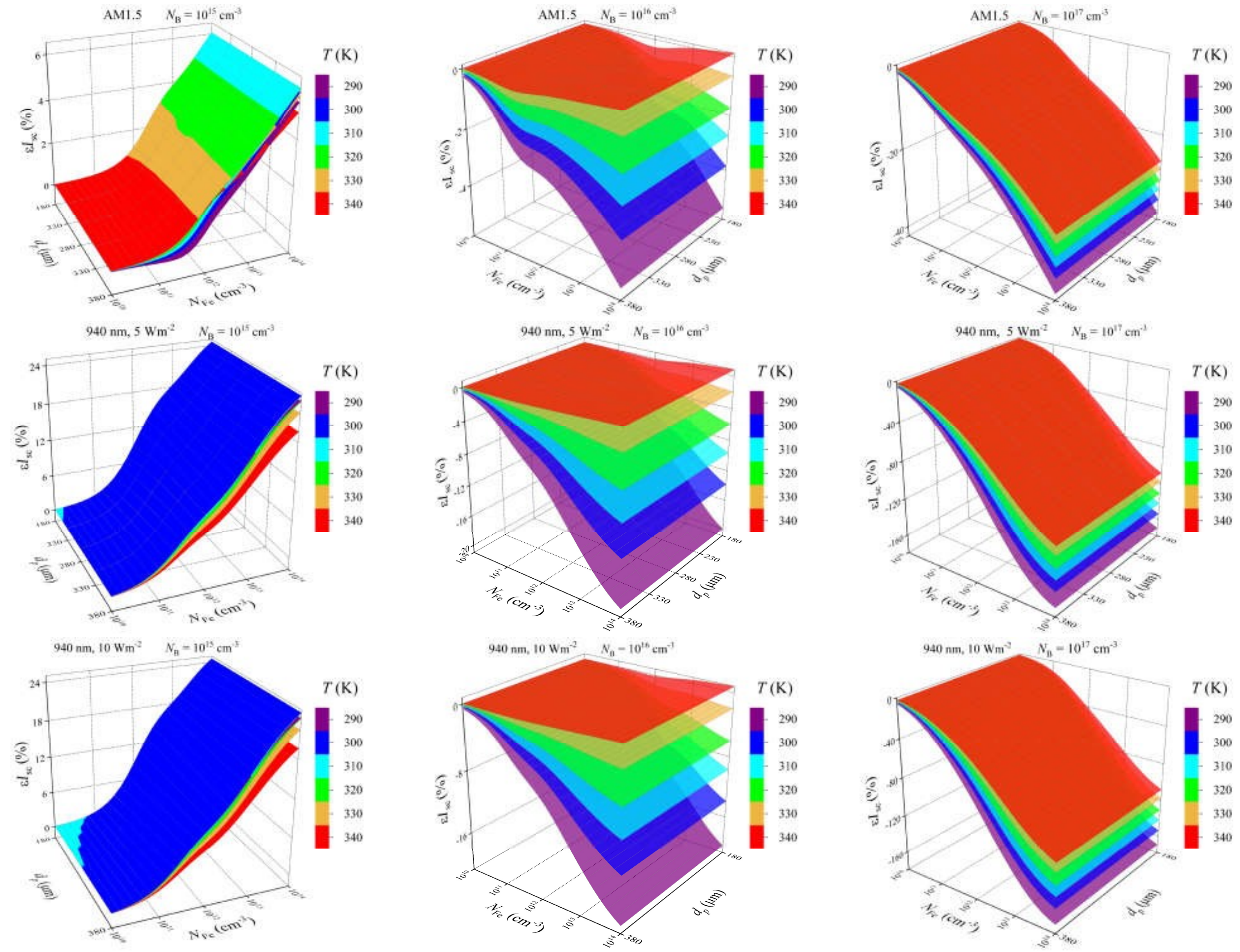

Fig.S3.5. Relative changes in short-circuit current caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and base depth for different temperatures. N_B, cm^{-3}: 10^{15} (left panels), 10^{16} (middle panels), 10^{17} (right panels).

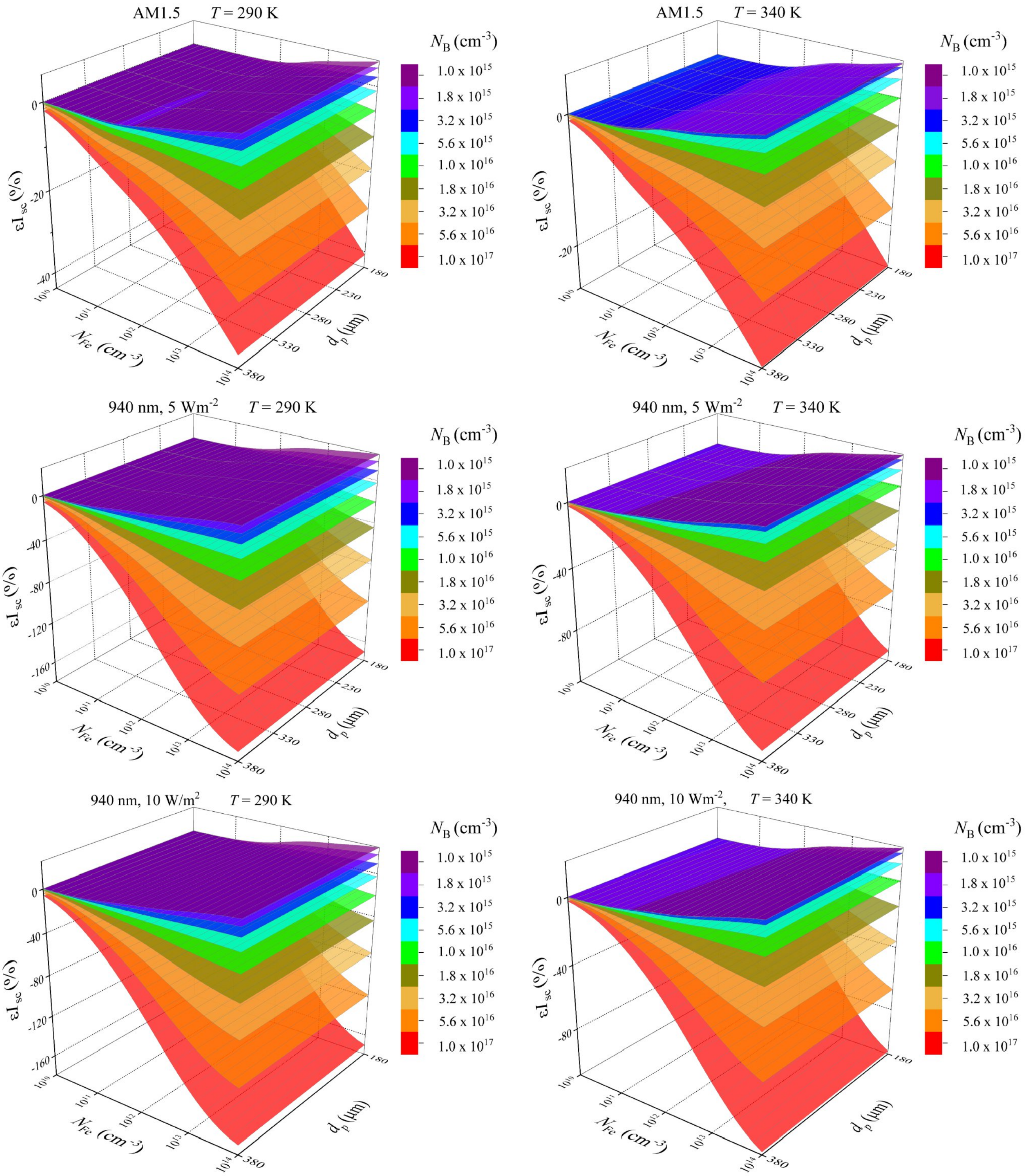

Fig.S3.6. Relative changes in short-circuit current caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and base depth for SC with different base doping level. T, K: 290 (left panels), 340 (right panels).

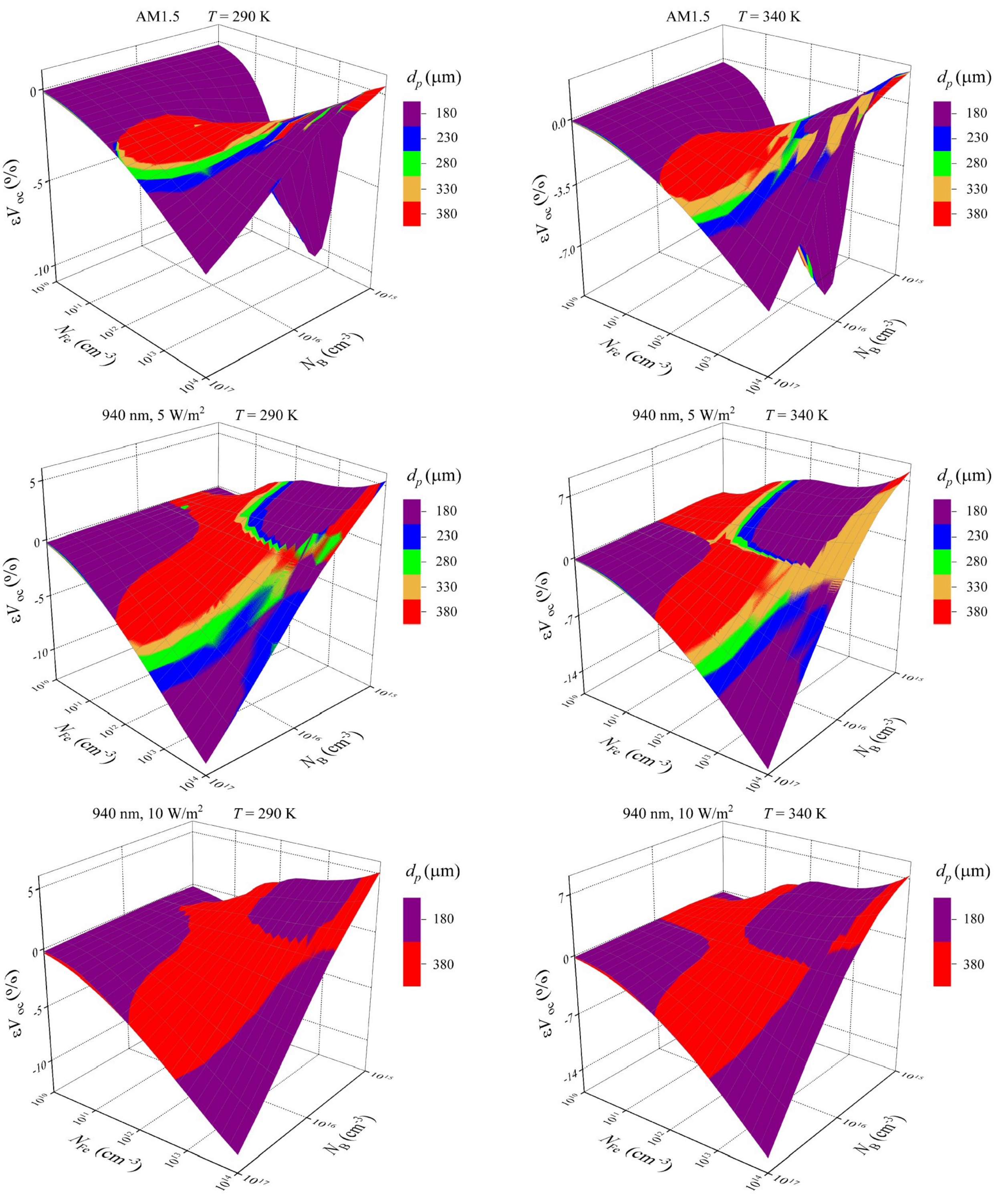

Fig.S3.7. Relative changes in open-circuit voltage caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level for SC with different base depth. T, K: 290 (left panels), 340 (right panels).

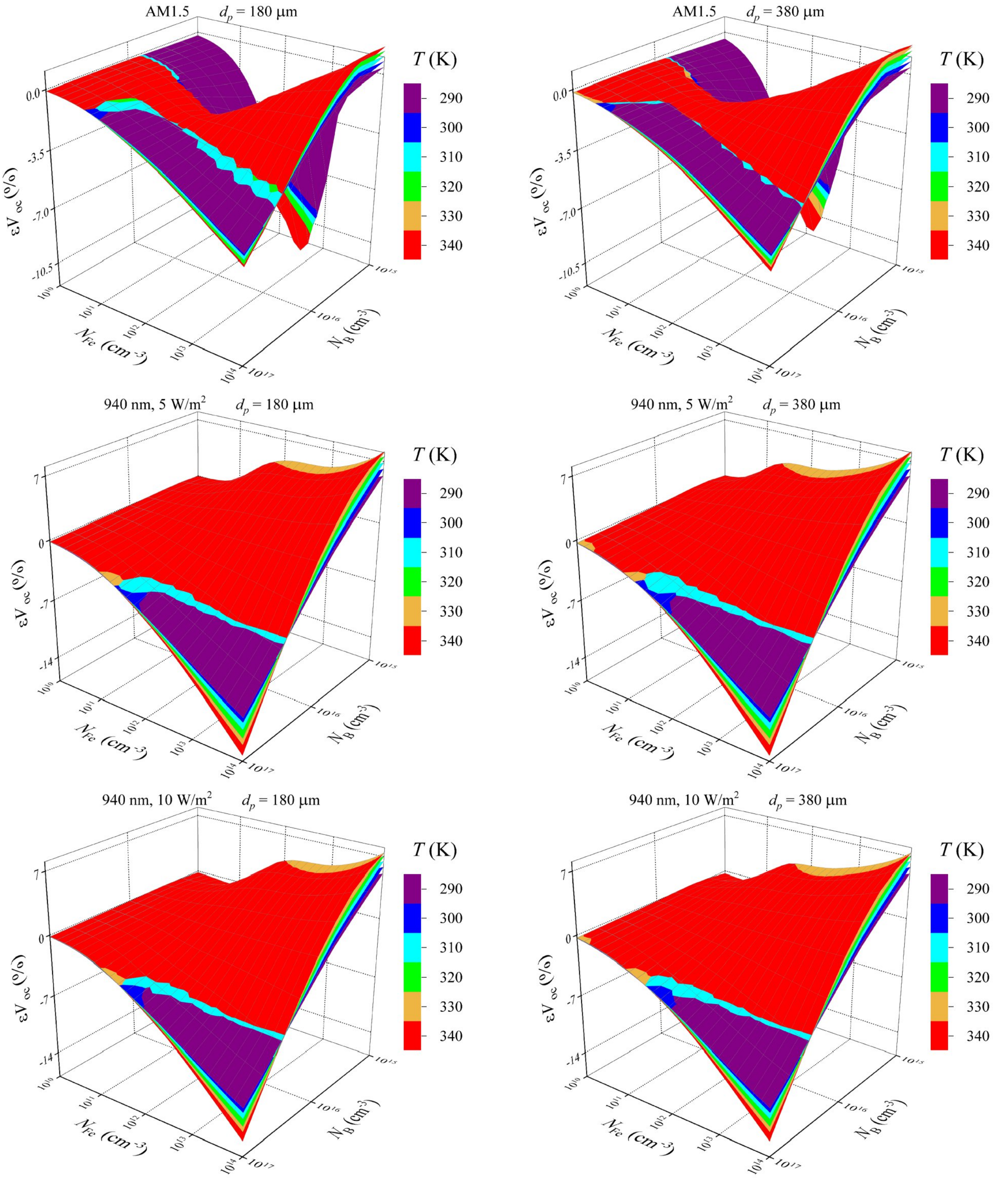

Fig.S3.8. Relative changes in open-circuit voltage caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level at different temperatures. d_p, µm: 180 (left panels), 380 (right panels).

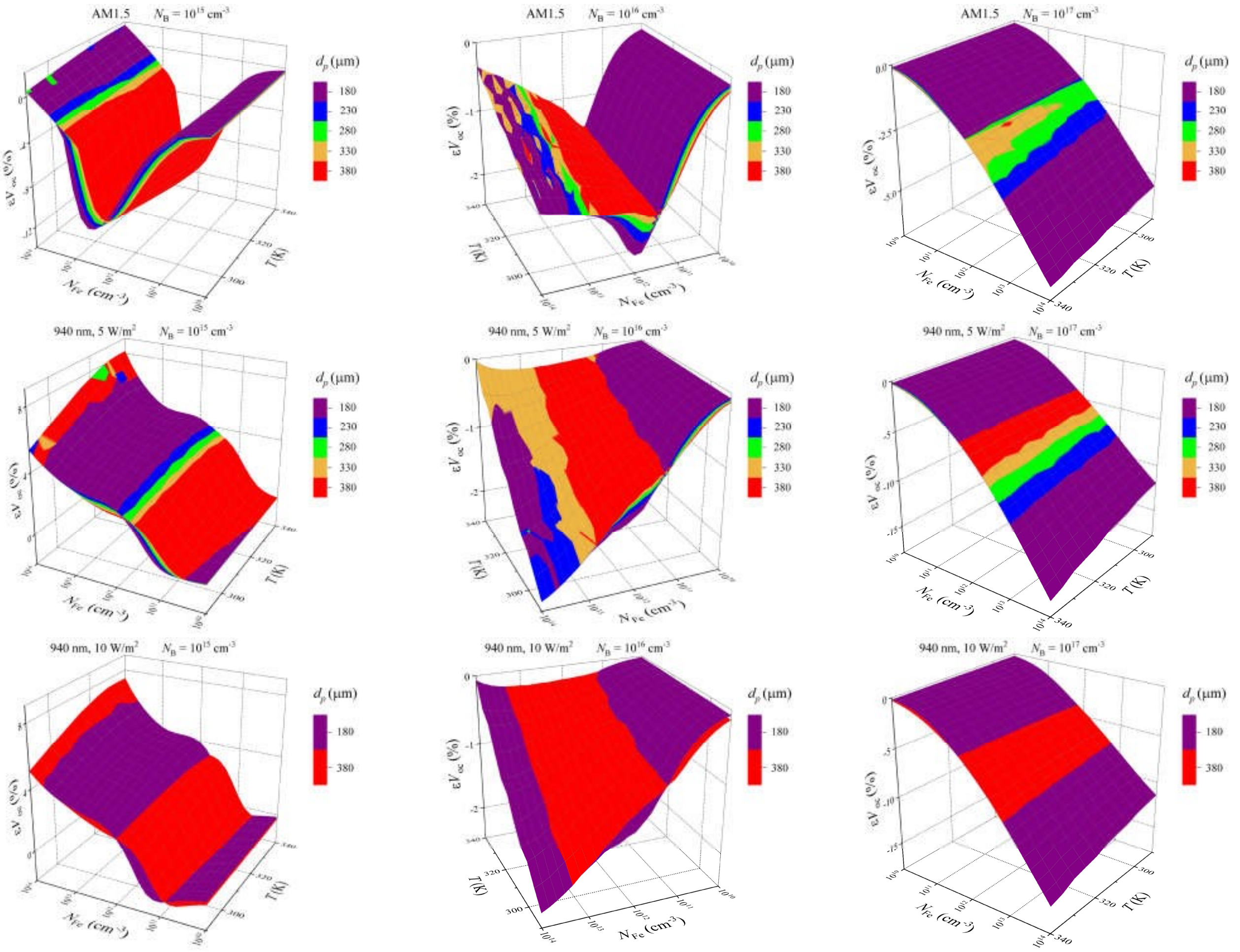

Fig.S3.9. Relative changes in open-circuit voltage caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and temperature for SC with different base depth. N_B, cm^{-3}: 10^{15} (left panels), 10^{16} (middle panels), 10^{17} (right panels).

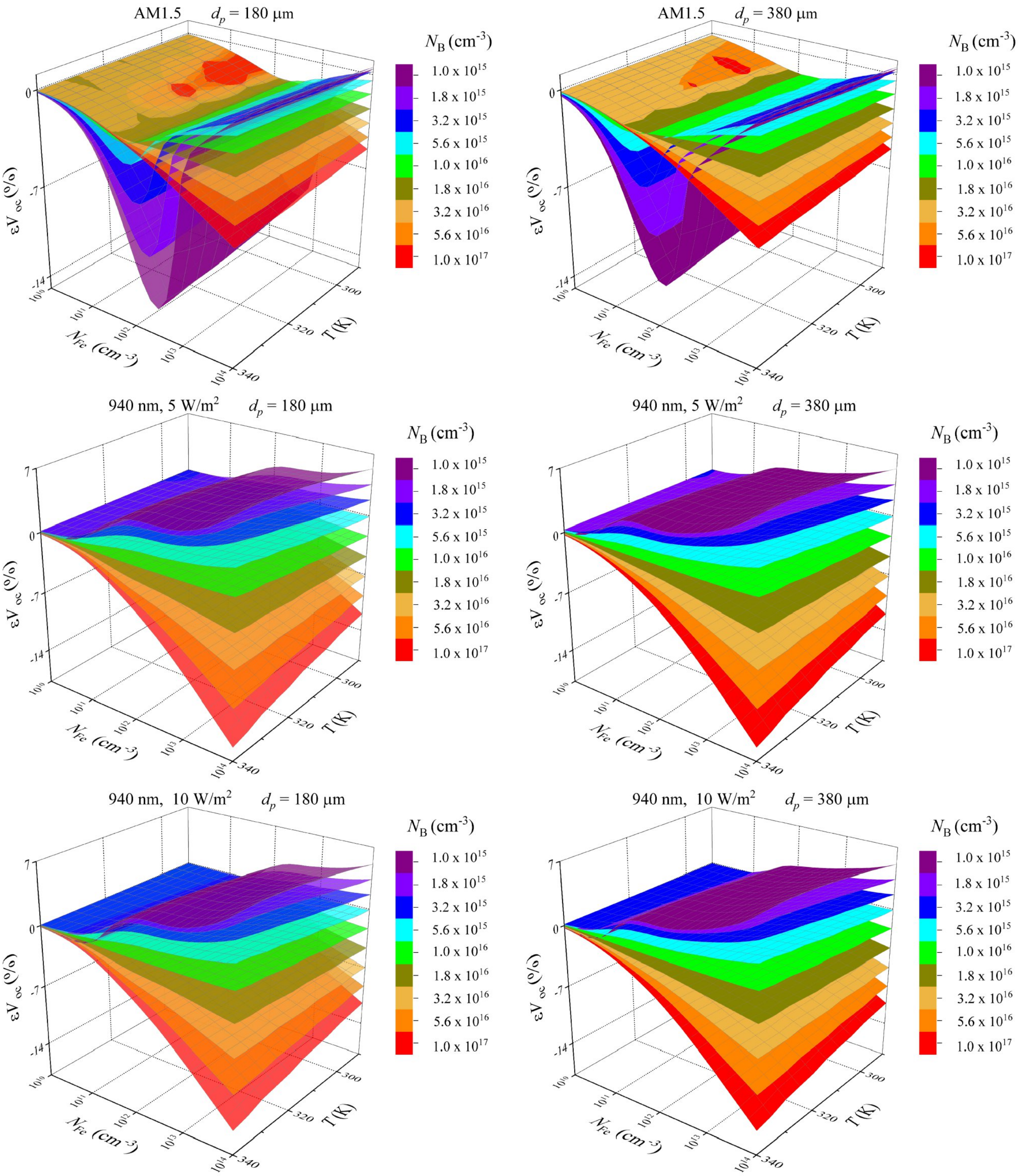

Fig.S3.10. Relative changes in open-circuit voltage caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and temperature for SC with different base doping level. d_p, μm: 180 (left panels), 380 (right panels).

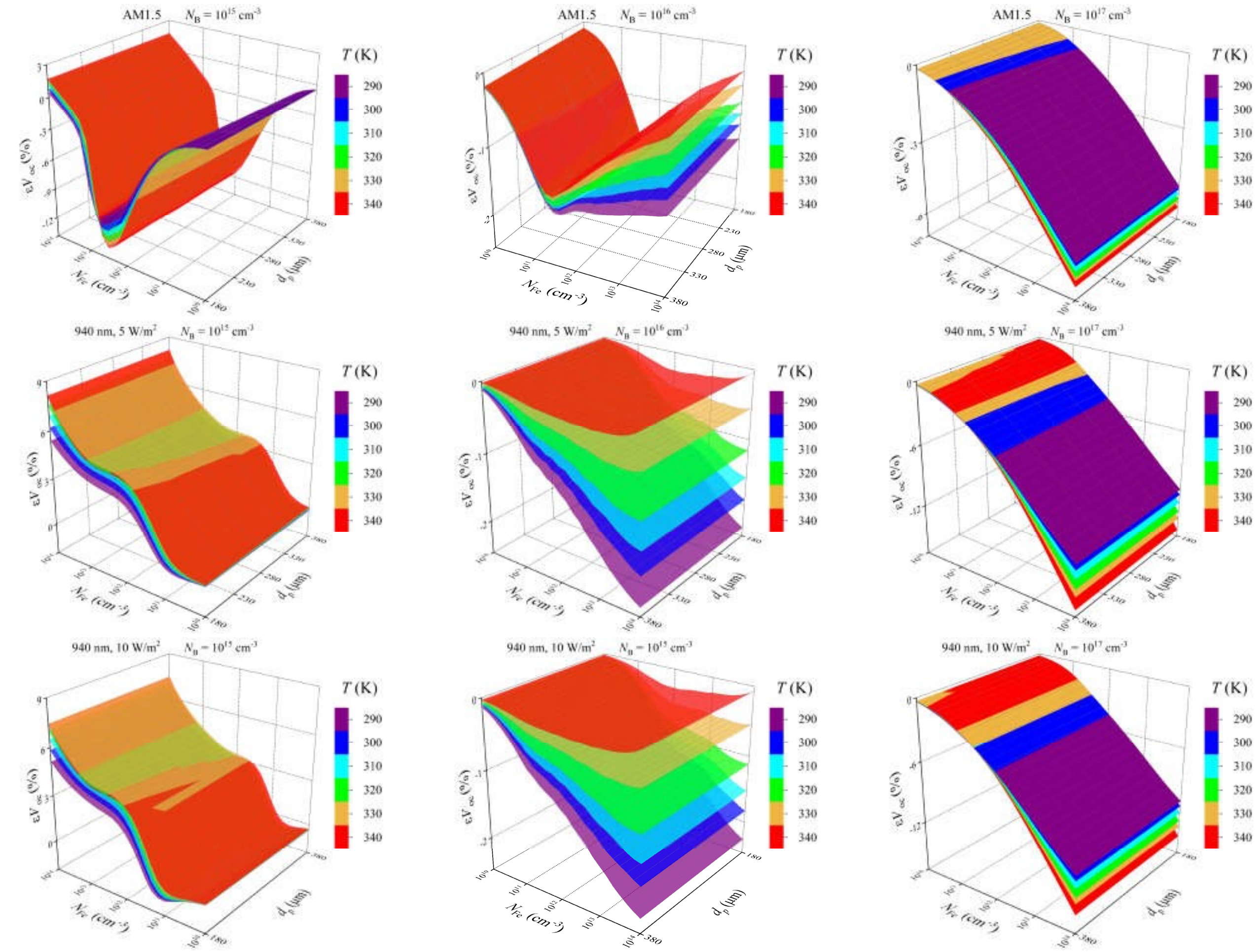

Fig.S3.11. Relative changes in open-circuit voltage caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and base depth for different temperatures. N_B, cm^{-3}: 10^{15} (left panels), 10^{16} (middle panels), 10^{17} (right panels).

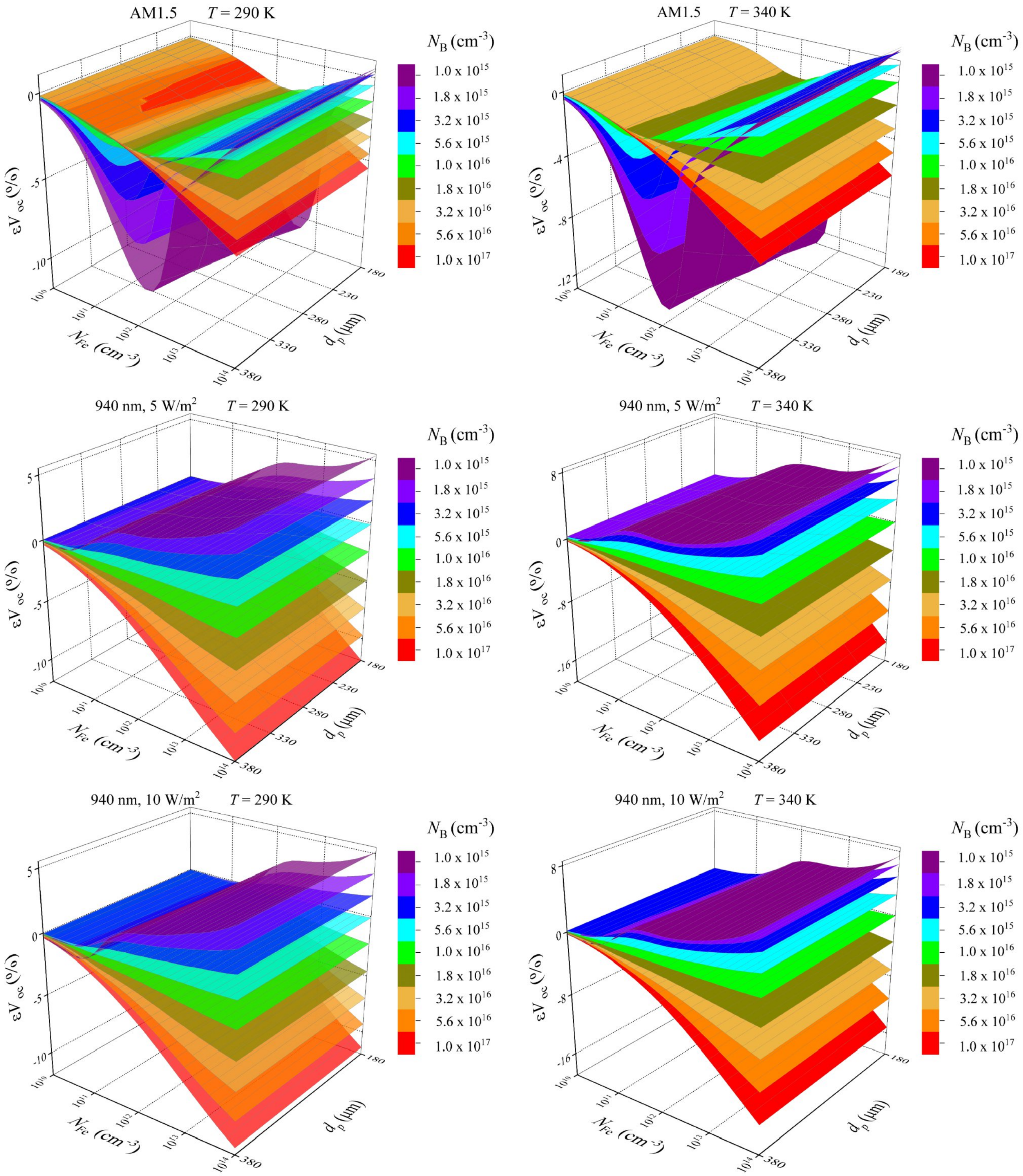

Fig.S3.12. Relative changes in open-circuit voltage caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and base depth for SC with different base doping level. T, K: 290 (left panels), 340 (right panels).

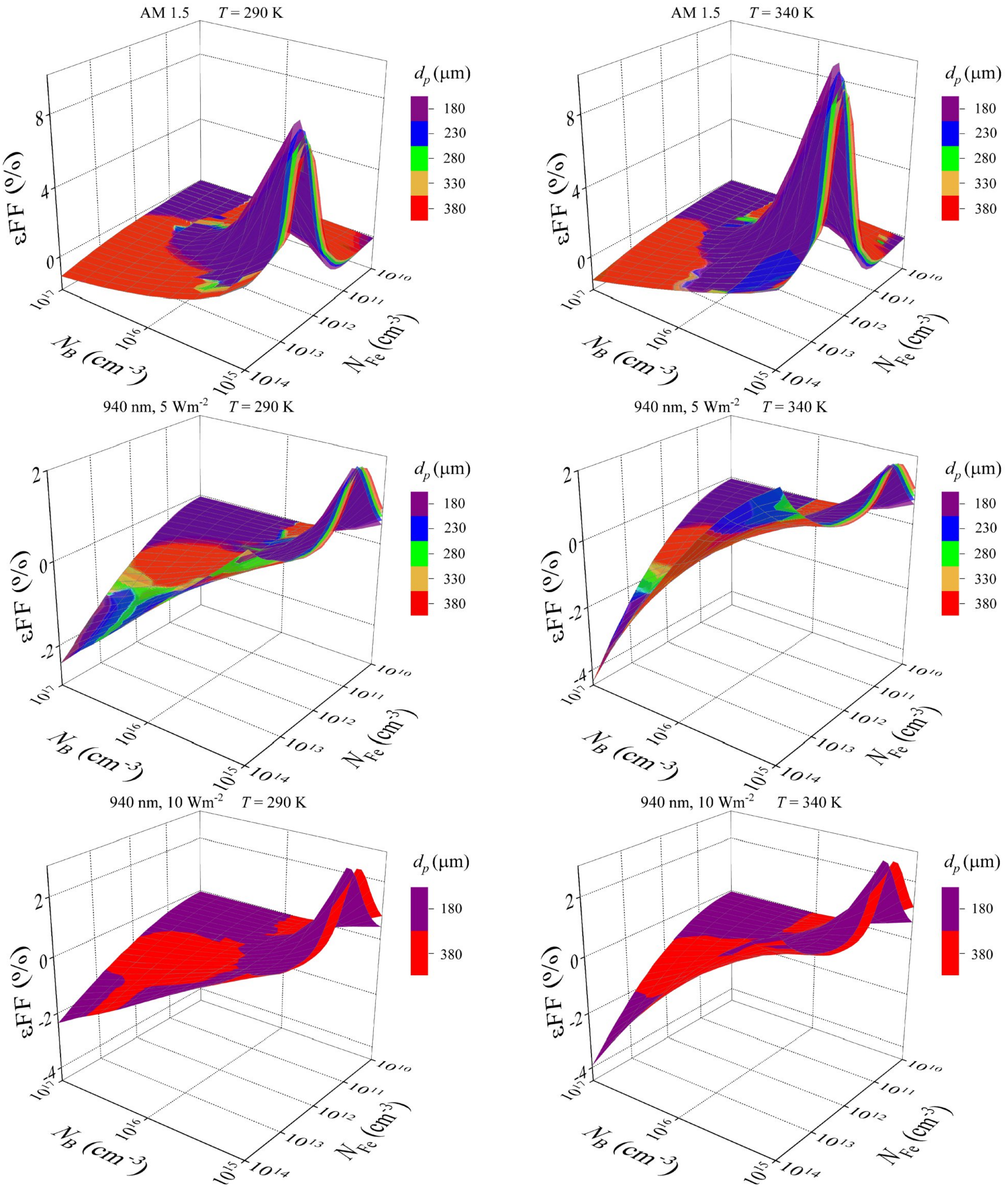

Fig.S3.13. Relative changes fill factor caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level for SC with different base depth. T, K: 290 (left panels), 340 (right panels).

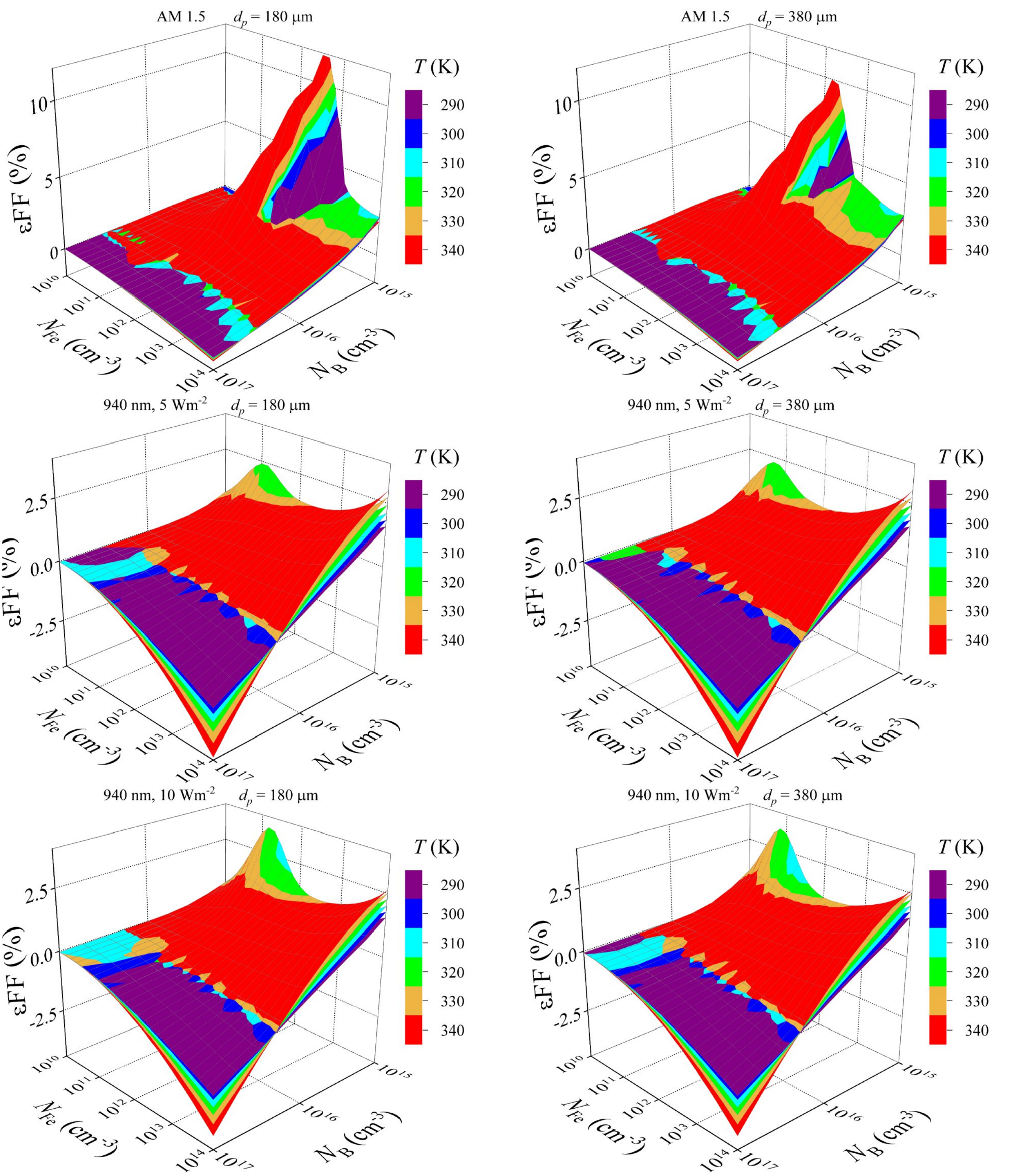

Fig.S3.14. Relative changes fill factor caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level at different temperatures. d_p, μm: 180 (left panels), 380 (right panels).

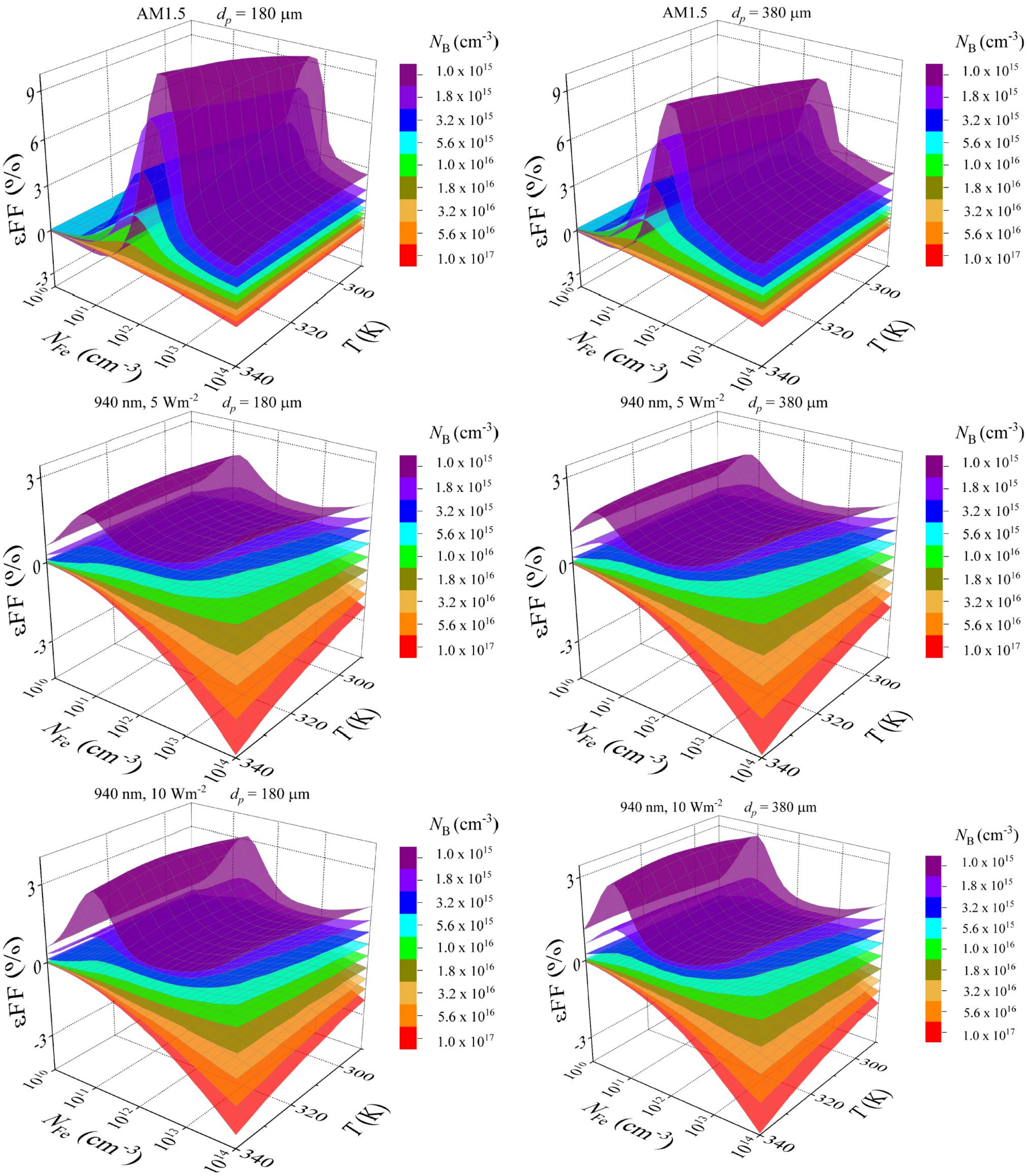

Fig.S3.15. Relative changes in fill factor caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and temperature for SC with different base doping level. d_p, μm: 180 (left panels), 380 (right panels).

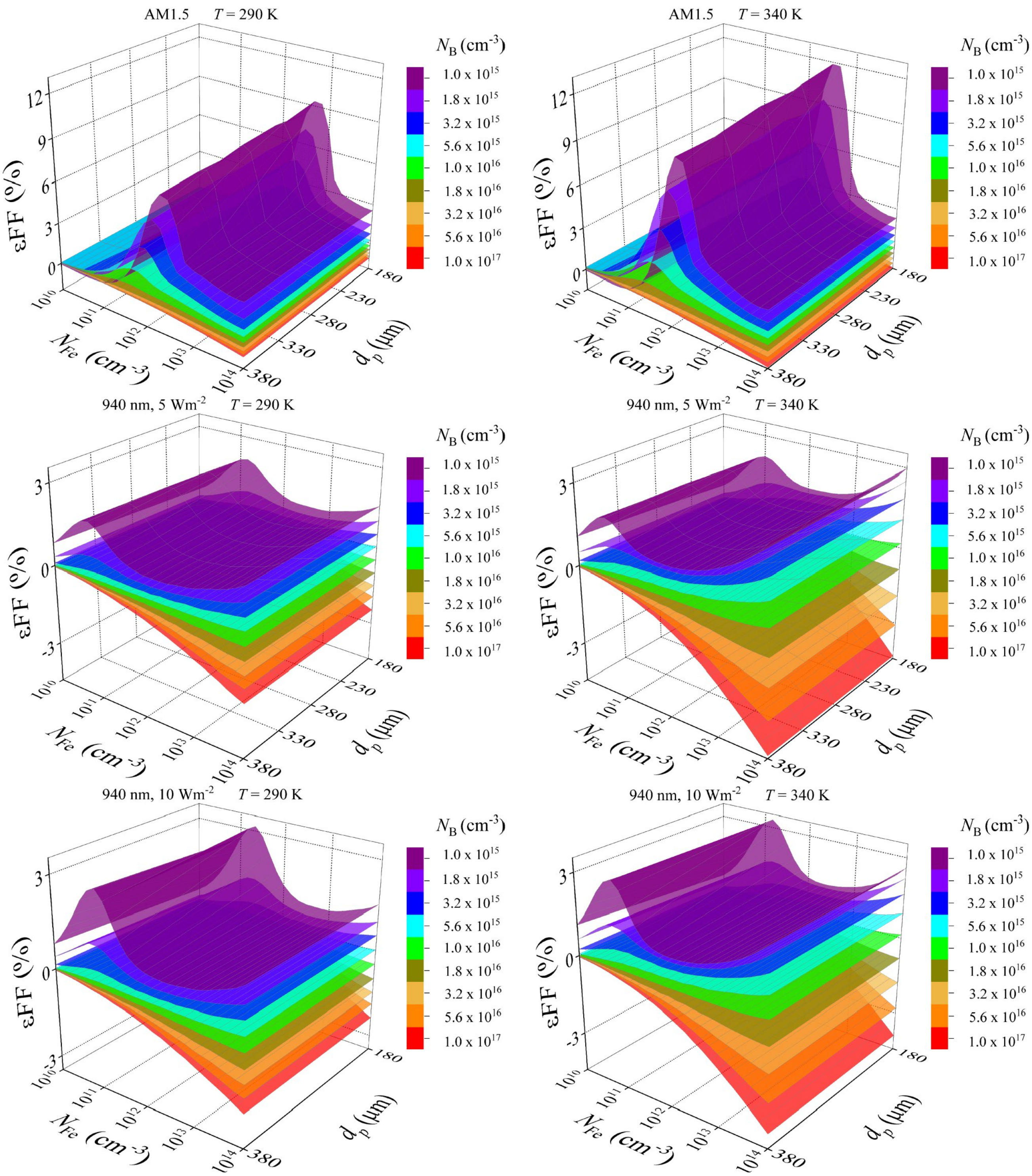

Fig.S3.16. Relative changes in fill factor caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and base depth for SC with different base doping level. T, K: 290 (left panels), 340 (right panels).

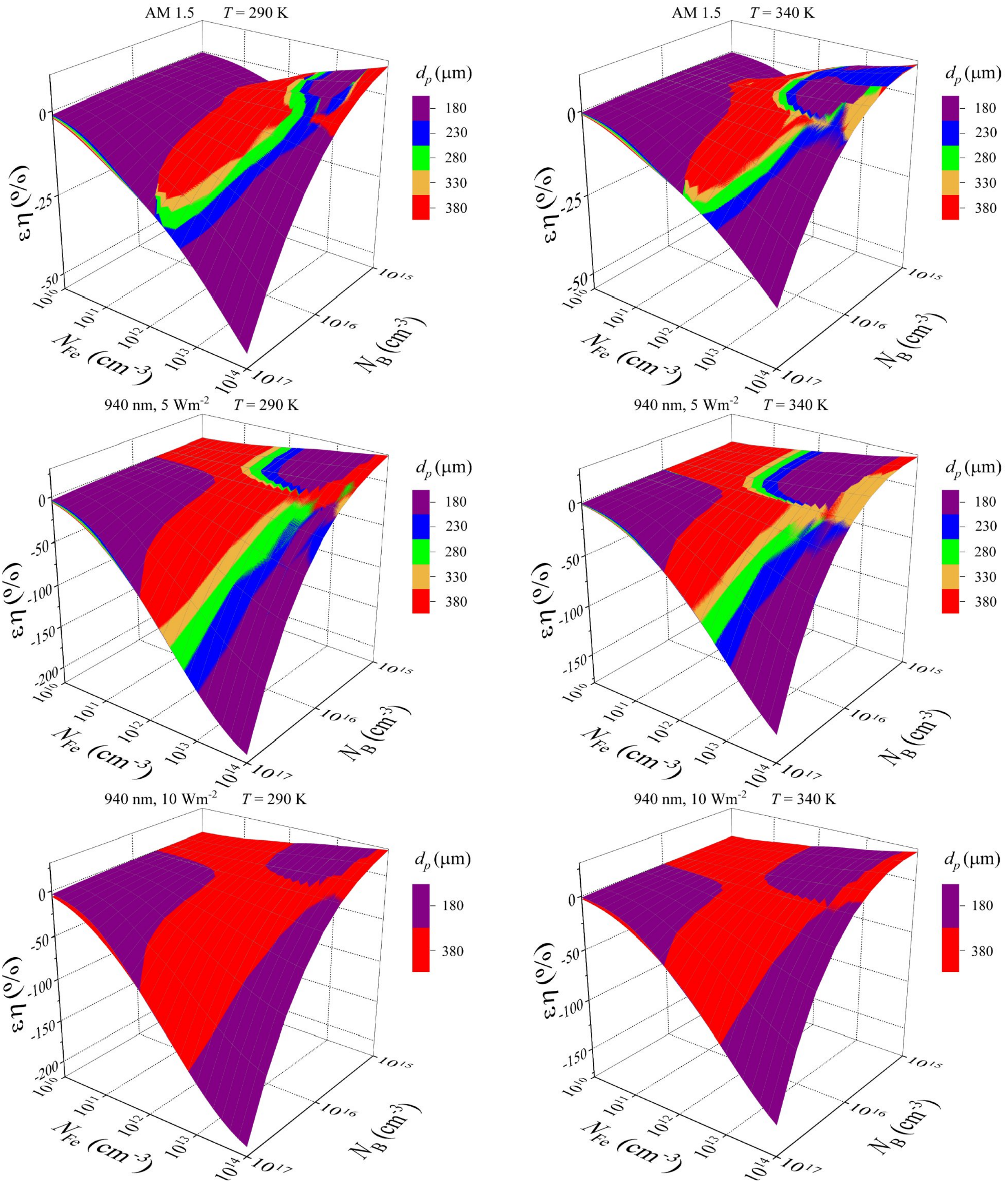

Fig.S3.17. Relative changes efficiency caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level for SC with different base depth. T, K: 290 (left panels), 340 (right panels).

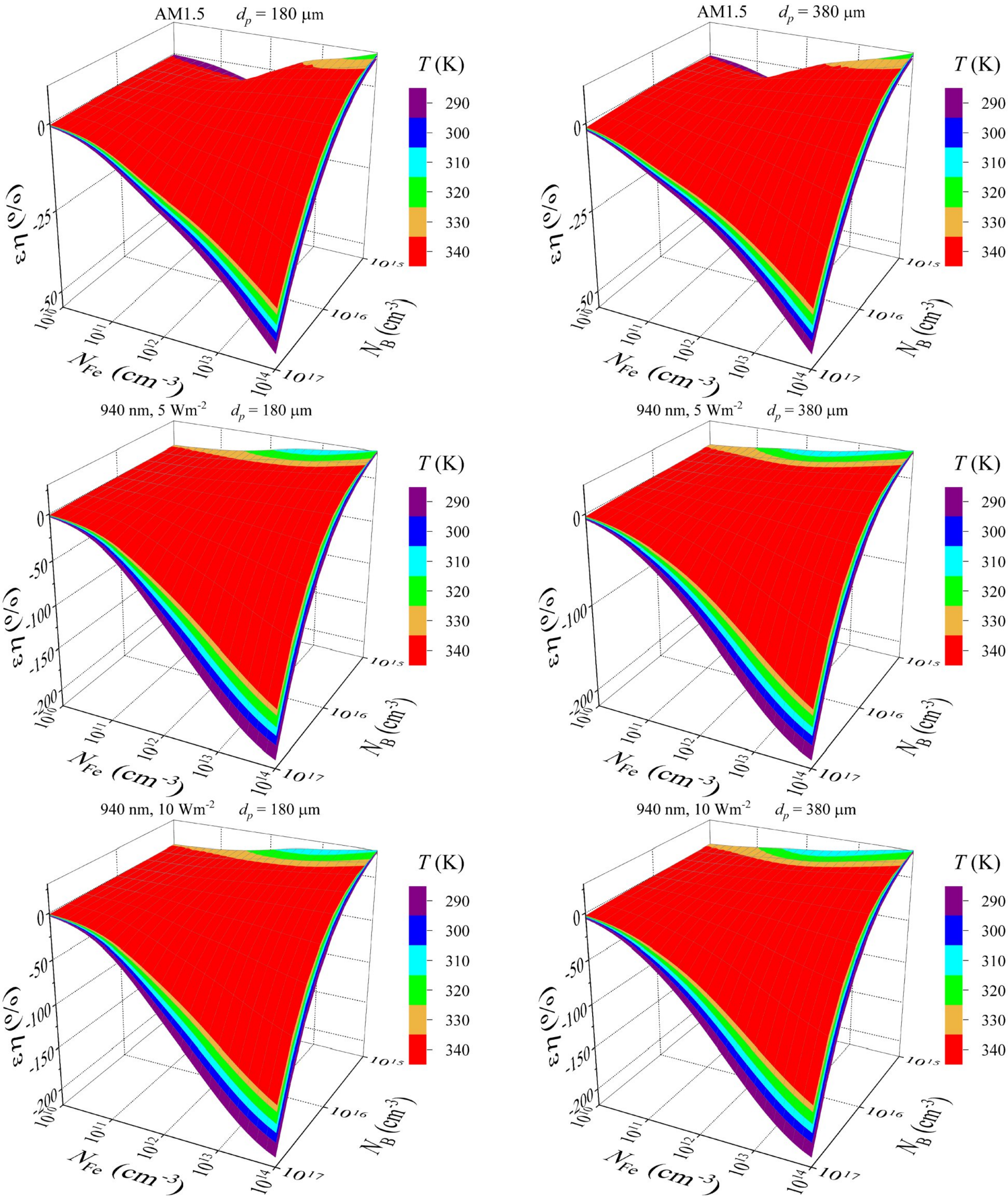

Fig.S3.18. Relative changes efficiency caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and doping level at different temperatures. d_p, μm: 180 (left panels), 380 (right panels).

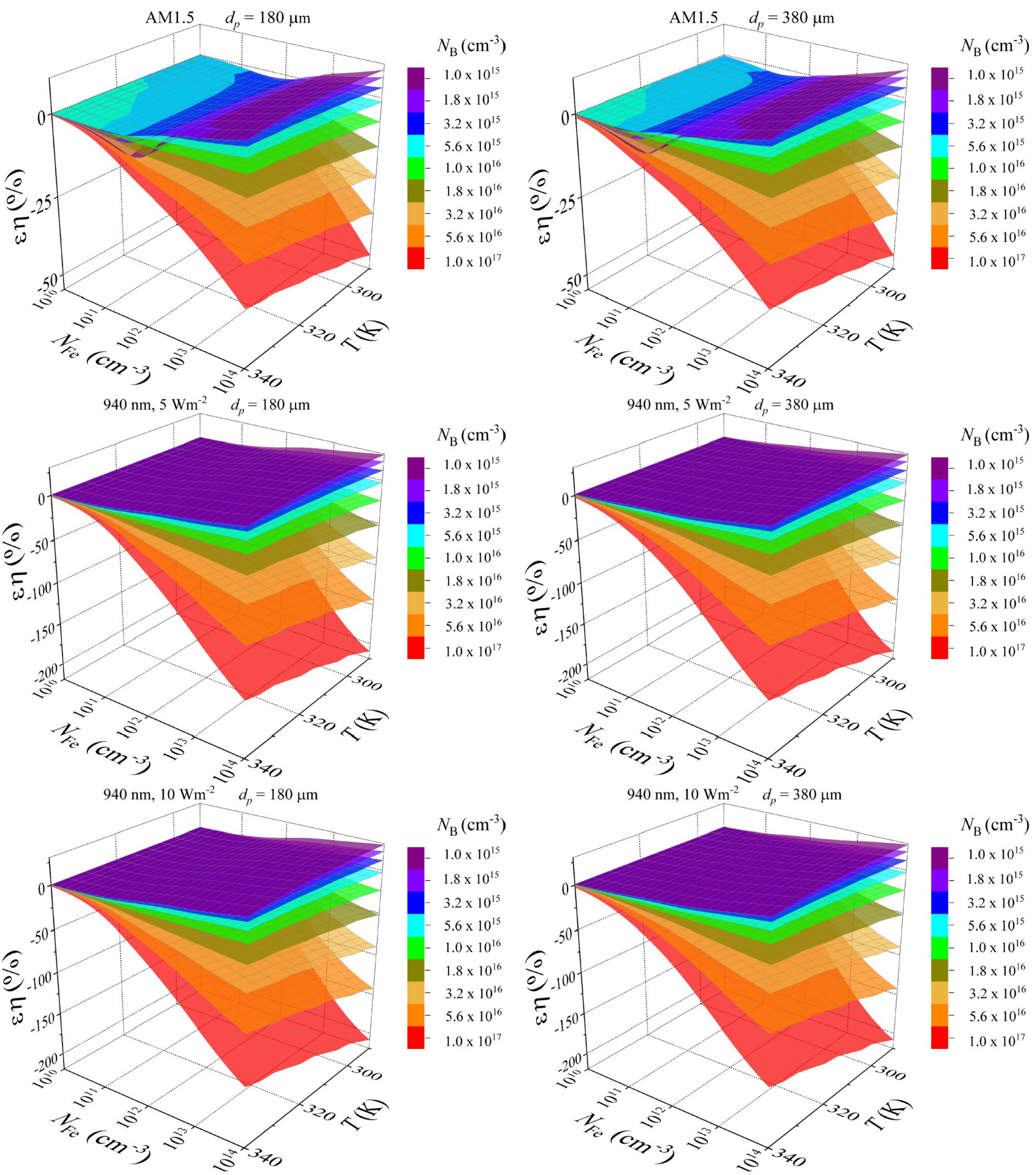

Fig.S3.19. Relative changes in efficiency caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and temperature for SC with different base doping level. d_p, μm: 180 (left panels), 380 (right panels).

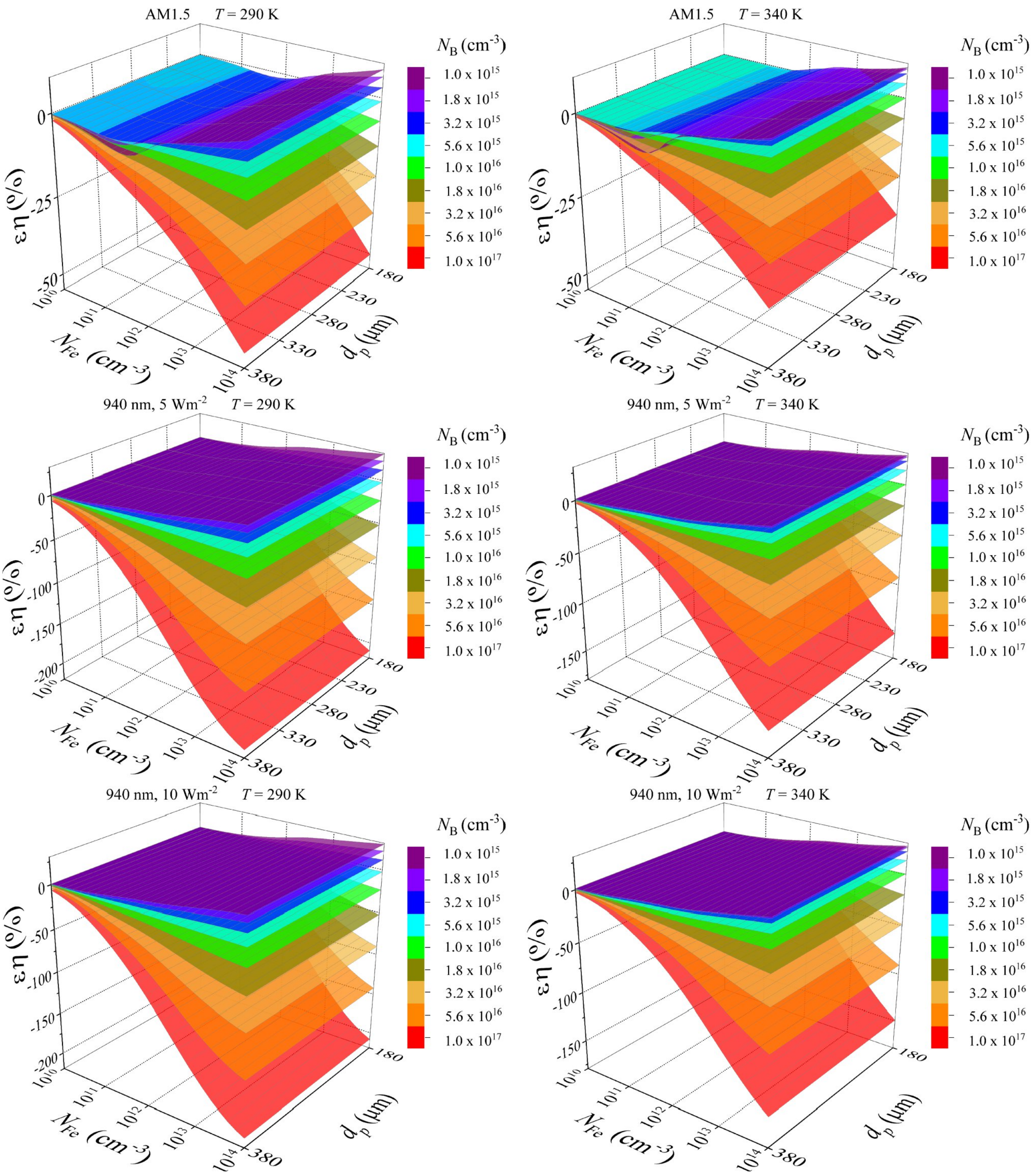

Fig.S3.20. Relative changes in efficiency caused by a complete dissociation of Fe_iB_s pairs as a function of iron concentration and base depth for SC with different base doping level. T, K: 290 (left panels), 340 (right panels).